

Relationship between Democracy & Human Development Indexes

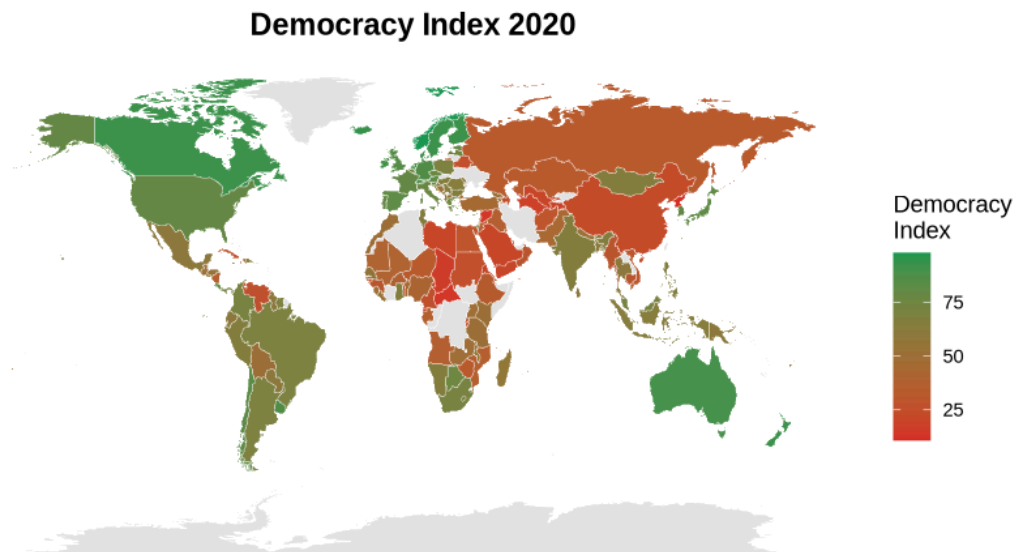
Brittany Rucker | October 2025

This report shows preliminary analysis on the relationship between the two indexes below. Both index datasets were pulled from the Gapminder data repository: <https://gapminder.org/data>.

- Democracy Index
 - Human Development Index
-

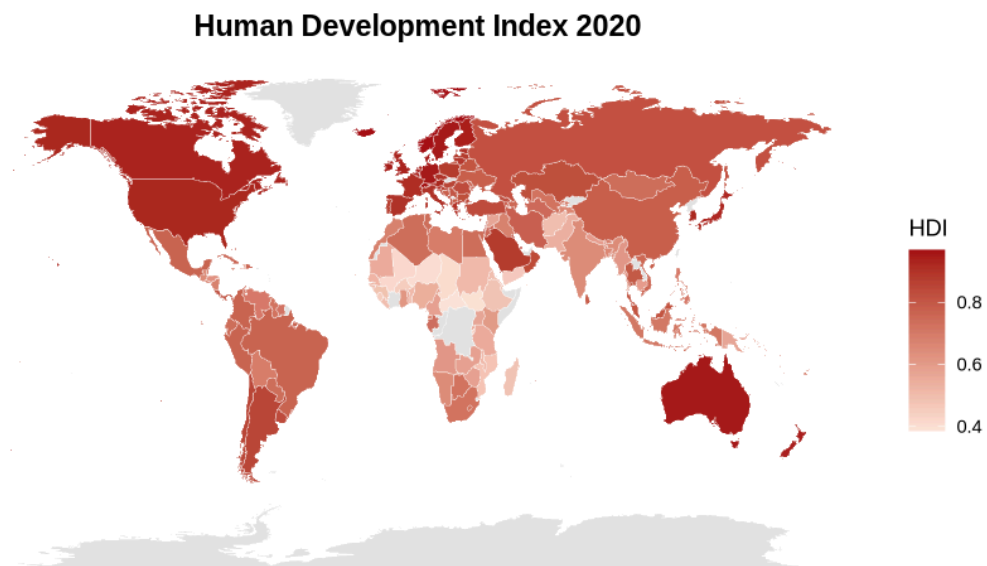
Democracy Index:

This map shows democracy levels across the world for year 2020, with green indicating higher democracy scores and red showing lower scores.



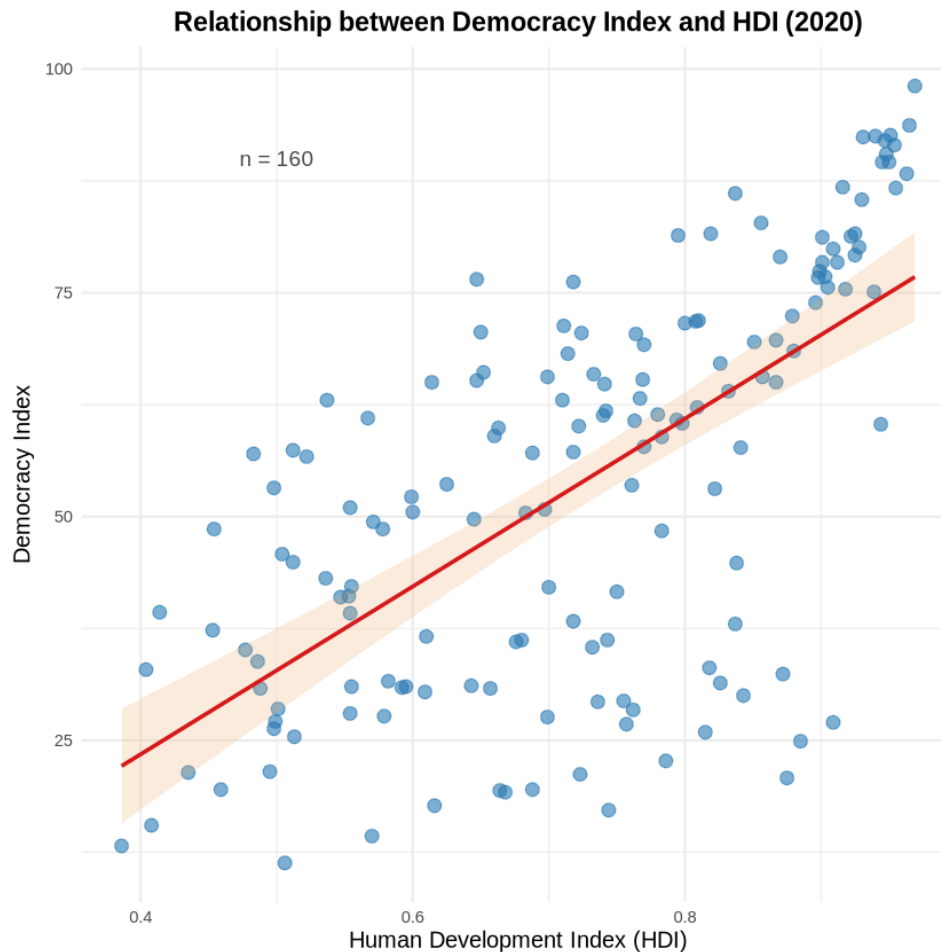
Human Development Index:

This map displays human development index (HDI) values globally for year 2020, with darker red representing higher human development levels.



Relational Analysis:

This scatterplot reveals a strong positive relationship between the two indicators, with a correlation of 0.651 and R^2 of 0.423. The blue line shows the linear trend, indicating that countries with higher human development tend to have stronger democratic institutions. However, there's considerable variation around the trend line, suggesting that while HDI is an important factor, it's not the only determinant of democracy levels.



Reflection:

The above analysis and plots were created the Julius ai() tool. Overall, I believe this tool is beneficial in creating top level analysis charts and data organization images. I like that the tool allows users the choice to choose from different ai-models to complete analysis. The program was super easy to use in terms of easy prompting. I would not rely on this tool to do more in depth data analysis.